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import os
import time
import streamlit as st
from google import genai
from google.genai import types

# Set up webpage configuration
st.set_page_config(
    page_title="Pro AI Video Generator",
    page_icon="🎬",
    layout="centered"
)

st.title("🎬 High-Precision AI Video Generator")
st.caption("Powered by Google Veo 3.1 (Image-to-Video & Text-to-Video)")

# Retrieve API key from Streamlit secrets or sidebar input
api_key = st.secrets.get("GEMINI_API_KEY") if "GEMINI_API_KEY" in st.secrets else None

if not api_key:
    api_key = st.sidebar.text_input("Gemini API Key:", type="password")

if not api_key:
    st.info("💡 Please enter your Gemini API Key in the sidebar to start.")
else:
    # Initialize the Google GenAI client
    client = genai.Client(api_key=api_key)

    # User inputs
    prompt = st.text_area(
        "Video Prompt / Instructions:",
        placeholder="Describe the movement or scene (e.g., 'A camera slow pan showing this character smiling gently')..."
    )

    # Image reference uploader
    uploaded_image = st.file_uploader(
        "Upload Reference Image (Optional for Image-to-Video):",
        type=["png", "jpg", "jpeg"]
    )

    if uploaded_image:
        st.image(uploaded_image, caption="Reference Image Preview", width=300)

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# Advanced Settings
st.subheader("Video Settings")
col1, col2, col3 = st.columns(3)
with col1:
    aspect_ratio = st.selectbox("Aspect Ratio", ["16:9", "9:16"])
with col2:
    duration = st.selectbox("Duration (sec)", [4, 6, 8])
with col3:
    resolution = st.selectbox("Resolution", ["720p", "1080p"])

negative_prompt = st.text_input(
    "Negative Prompt (Things to avoid):",
    value="glitches, distortion, blurry, low quality, unnatural movement, flickering"
)

# Generation Trigger
if st.button("Generate Video 🚀", use_container_width=True, type="primary"):
    if not prompt.strip() and not uploaded_image:
        st.warning("Please enter a prompt or upload a reference image.")
    else:
        with st.spinner("Processing generation request... (This takes ~1 to 2 minutes)"):
            try:
                # Prepare image reference if uploaded
                image_input = None
                if uploaded_image:
                    image_bytes = uploaded_image.getvalue()
                    image_input = types.Image(
                        image_bytes=image_bytes,
                        mime_type=uploaded_image.type
                    )

                # Configure video params
                config = types.GenerateVideosConfig(
                    aspect_ratio=aspect_ratio,
                    duration_seconds=duration,
                    resolution=resolution,
                    negative_prompt=negative_prompt,
                    number_of_videos=1,
                )

                # Call Veo 3.1 model
                operation = client.models.generate_videos(
                    model="veo-3.1-generate-001",
                    prompt=prompt if prompt.strip() else None,

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        image=image_input,
        config=config,
    )

    # Poll until generation completes
    while not operation.done:
        time.sleep(10)
        operation = client.operations.get(operation)

    # Retrieve result
    result = operation.result
    generated_video = result.generated_videos[0]
    video_uri = generated_video.video.uri

    st.success("🎉 Video generated successfully!")
    st.video(video_uri)

except Exception as e:
    st.error(f"Generation error: {e}")

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